

19.3 Chiral Symmetry Breaking and Life

The chemistry of life as we know it is founded on a remarkable asymmetry. A molecule whose geometrical structure is not identical to its mirror image is said to possess **chirality**, or handedness. Mirror-image structures of a chiral molecule are called **enantiomers**. Just as we distinguish the left and the right hand, the two mirror-image structures are identified as L- and D-enantiomers (L for “levo” and D for “dextro”; *R* and *S* is another convention of identifying the two enantiomers). Amino acids, the building blocks of proteins, and deoxyribose in DNA are chiral molecules. From bacteria to man, nearly all amino acids that take part in the chemistry of life are L-amino acids (Fig. 19.2) and the riboses in DNA and RNA are D-ribose (Fig. 19.3). As Francis Crick notes, “The first great unifying principle of biochemistry is that the key molecules have the same hand in all organisms.” This is all the more remarkable because chemical reactions show equal preference for the two mirror-image forms (except for very small differences due to parity-nonconserving electroweak interactions [4–6]).

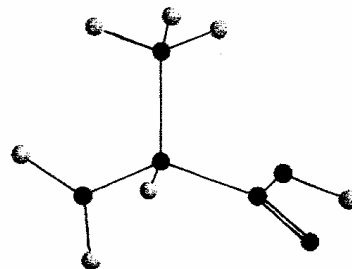
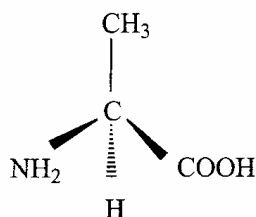


Figure 19.2 Proteins are made exclusively of L-amino acids. The amino acid shown is L-alanine. In other L-amino acids, different groups of atoms take the place of CH_3

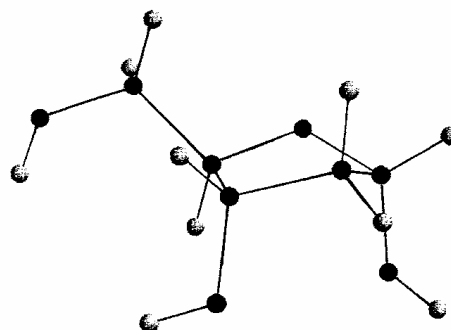
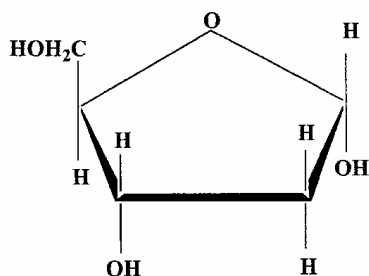
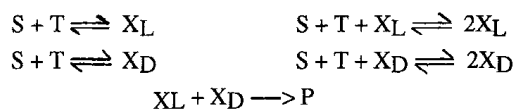


Figure 19.3 2-deoxy-D-ribose shown above is a basic chiral building block of DNA. Its mirror image structure, 2-deoxy-L-ribose, is excluded from the chemistry of life

Biochemistry's hidden asymmetry was discovered by Louis Pasteur in 1857. Nearly 150 years later, its true origin remains an unsolved problem, but we can see how such a state might be realized in the framework of dissipative structures. First, we note that such an asymmetry can arise only under far-from-equilibrium conditions; at equilibrium the concentrations of the two enantiomers will be equal. The maintenance of this asymmetry requires constant catalytic production of the preferred enantiomer in the face of interconversion between enantiomers, called **racemization**. (Racemization drives the system to the equilibrium state in which the concentrations of the two enantiomers will become equal.) Second, following the paradigm of order through fluctuations, we will presently see how, in systems with appropriate chiral autocatalysis, the thermodynamic branch, which contains equal amounts of L- and D-enantiomers, can become unstable. The instability is accompanied by the bifurcation of asymmetric states, or states of broken symmetry, in which one enantiomer dominates. Driven by random fluctuations, the system makes a transition to one of the two possible states.

In 1953 F. C. Frank [7] devised a simple model reaction scheme with chiral autocatalysis that could amplify a small initial asymmetry. We shall modify this

Reaction scheme:



$$\alpha = ([X_L] - [X_D])/2$$

$$\lambda = [S][T]$$

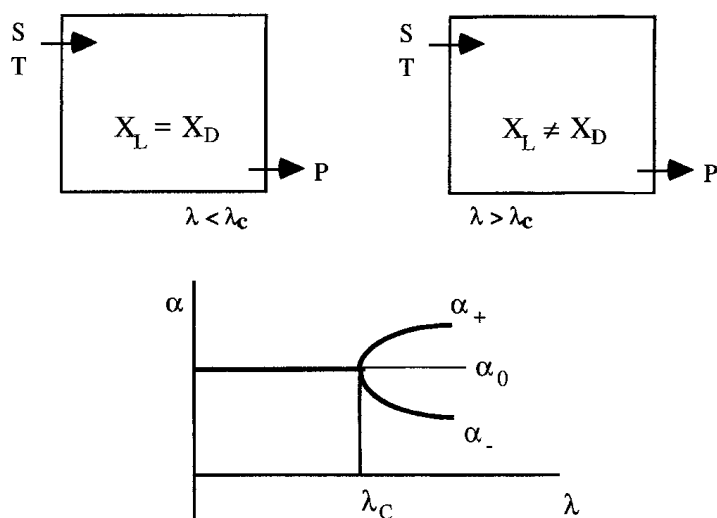


Figure 19.4 A simple autocatalytic reaction scheme in which X_L and X_D are produced with equal preference. However, in an open system, this leads to a dissipative structure in which $X_L \neq X_D$, a state of broken symmetry. A bifurcation diagram shows some general features of transitions to dissipative structures

reaction scheme so that its nonequilibrium aspects, instability and bifurcation of symmetry breaking states can be clearly seen (Fig. 19.4). It includes chirally autocatalytic reactions:



Each enantiomer of X is produced directly from the achiral* reactants S and T, as shown in (19.3.1) and (19.3.3) and autocatalytically, as shown in (19.3.2) and (19.3.4). In addition, the two enantiomers react with one another and turn into an inactive compound, P. Due to symmetry, the rate constants for the direct reactions, (19.3.1) and (19.3.3), as well as the autocatalytic reactions, (19.3.2) and (19.3.4), must be equal. It is easy to see that at equilibrium the system will be in a *symmetric state*, i.e. $[X_L] = [X_D]$ (exc. 19.3). Now let us consider an open system into which S and T are pumped and from which P is removed. For mathematical simplicity, we assume that the pumping is done in such a way that the concentrations [S] and [T] are maintained at a fixed level, and that due to removal of P the reverse reaction in (19.3.5) may be ignored. The kinetic equation, of this system are

$$\frac{d[X_L]}{dt} = k_{1f}[S][T] - k_{1r}[X_L] + k_{2f}[X_L][S][T] - k_{2r}[X_L]^2 - k_3[X_L][X_D] \quad (19.3.6)$$

$$\frac{d[X_D]}{dt} = k_{1f}[S][T] - k_{1r}[X_D] + k_{2f}[X_D][S][T] - k_{2r}[X_D]^2 - k_3[X_L][X_D] \quad (19.3.7)$$

To make the symmetric and asymmetric states explicit, it is convenient to define the following variables:

$$\lambda = [S][T] \quad \alpha = \frac{[X_L] - [X_D]}{2} \quad \beta = \frac{[X_L] + [X_D]}{2} \quad (19.3.8)$$

When equations (19.3.6) and (19.3.7) are rewritten in terms of α and β (exc. 19.4), we have

$$\frac{d\alpha}{dt} = -k_{1r}\alpha + k_{2f}\lambda\alpha - 2k_{2r}\alpha\beta \quad (19.3.9)$$

$$\frac{d\beta}{dt} = k_{1f}\lambda - k_{1r}\beta + k_{2f}\lambda\beta - k_{2r}(\beta^2 + \alpha^2) - k_3(\beta^2 - \alpha^2) \quad (19.3.10)$$

The stationary states of this equation can be obtained after a little calculation (by setting $d\alpha/dt = d\beta/dt = 0$). A complete analytic study of the solutions of (19.3.9) and (19.3.10) and their stability is somewhat lengthy and it can be found in the literature [8]. We shall only state the main results of this analysis. With the Mathematica code provided in Appendix 19.1, the reader can explore the properties of the equation quite easily and verify the phenomenon of chiral symmetry breaking in this system (Fig. 19.5).

* Objects that do not possess a sense of handedness are called achiral. The molecule NH_3 is an example of an achiral molecule.

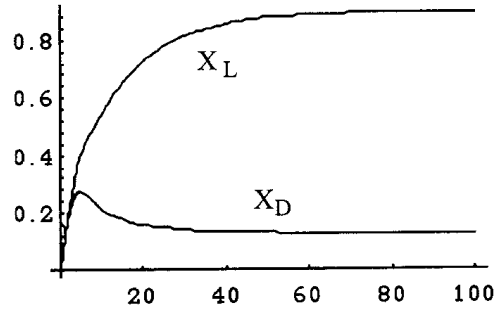


Figure 19.5 Time evolution of X_L and X_D obtained using Mathematica code A given in Appendix 19.1. For $\lambda > \lambda_c$ a small initial fluctuation in X_L grows to establish a state of broken symmetry in which the concentrations of X_L and X_D are unequal

For small values of λ the stable stationary state is

$$\alpha_s = 0 \quad (19.3.11)$$

$$\beta_s = \frac{2k_{2r}\beta_a + \sqrt{(2k_{2r}\beta_a)^2 + 4(k_{2r} + k_3)k_{1f}\lambda}}{2(k_{2r} + k_3)} \quad (19.3.12)$$

in which

$$\beta_a = \frac{k_{2f}\lambda - k_{1r}}{2k_{2r}}$$

This is a *symmetric* solution in which $[X_L] = [X_D]$ (as indicated by the subscript s). Using the stability analysis described in the previous chapter, it can be shown that this symmetric solution becomes unstable when λ greater than a critical value λ_c . The value of λ_c is given by

$$\lambda_c = \frac{s + \sqrt{s^2 - 4k_{2f}^2 k_{1r}^2}}{2k_{2f}^2} \quad (19.3.13)$$

where

$$s = 2k_{2f}k_{1r} + \frac{4k_{2r}^2 k_{1r}}{k_3 - k_{2r}} \quad (19.3.14)$$

For the system of equations (19.3.9) and (19.3.10) it is possible to obtain an asymmetric stationary solution analytically:

$$\alpha_a = \pm \sqrt{\beta_a^2 - \frac{k_{1f}\lambda}{k_3 - k_{2r}}} \quad (19.3.15)$$

$$\beta_a = \frac{k_{2f}\lambda - k_{1r}}{2k_{2r}} \quad (19.3.16)$$

in which the subscript *a* stands for *asymmetric*. We recommend the reader to use Mathematica code A in Appendix 19.1 to verify all these properties of the system.

To date, no chemical reaction has produced chiral asymmetry in this simple manner. However, symmetry breaking does occur in the crystallization of NaClO₃ [9, 10] in far from equilibrium conditions. The simple model however leads to interesting conclusions regarding the sensitivity of bifurcation discussed below.

NONEQUILIBRIUM SYMMETRY BREAKING AND THE ORIGIN OF BIOMOLECULAR ASYMMETRY

The above example shows how a far-from-equilibrium chemical system can generate and maintain chiral asymmetry, but it only provides a general framework in which we must seek the origins of biomolecular handedness. The origin of biomolecular handedness, or life's **homochirality**, remains to be explained [11, 12]. Here we shall confine our discussion to how the theory of nonequilibrium symmetry breaking contributes to this important topic. We cannot yet say with confidence whether chiral asymmetry arose in a prebiotic (i.e. before life) process and facilitated the evolution of life, or whether some primitive form of life that incorporated both L- and D-amino acids arose first and subsequent evolution of this life form led to the homochirality of L-amino acids and D-sugars. Both views have their proponents.

A related question is whether the dominance of L-amino acids in biochemistry was a matter of chance or whether it was a consequence of the extremely small but systematic chiral asymmetry due to electroweak interactions that is known to exist at the atomic and molecular levels [4–6, 13, 14]. Theories that support both views have been put forward but there is no general consensus on this matter either, mainly because there is a dearth of persuasive experimental evidence. However, the theory of nonequilibrium symmetry breaking provides a valuable means of assessing the plausible role of different models. For example, if we consider a prebiotic symmetry-breaking process that might have occurred in the oceans, it is possible to develop a general theory of symmetry breaking. A parameter λ , similar to the one in the above model, can be defined for any

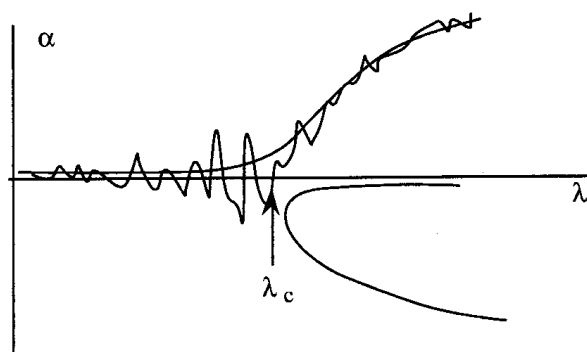


Figure 19.6 A symmetry-breaking transition or bifurcation in the presence of a small bias that favors one of the bifurcating branches. It can be analyzed through the general equation (19.3.17) and the probability of the system making a transition to the favored branch is given by equation (19.3.18)

symmetry-breaking system. When $\lambda < \lambda_c$, the system will be in a symmetric state; for $\lambda > \lambda_c$ the symmetric state will become unstable and evolve to an asymmetric state. Furthermore, regardless of the complexities of the reaction scheme, based on symmetry considerations only, it is possible to describe the bifurcation of the chiral symmetry-breaking states with an equation of the following type [15, 16]:

$$\frac{d\alpha}{dt} = -A\alpha^3 + B(\lambda - \lambda_c)\alpha + Cg + \sqrt{\varepsilon}f(t) \quad (19.3.17)$$

in which the coefficients A , B , and C depend on the concentrations of the reactants and on the reaction rates (Fig. 19.6). The parameter g , is a small systematic bias, such as due to the electroweak force [5, 14] or other systematic chiral influences such as spin-polarized electrons that emerge from radioactive decay [17], or circularity polarized electromagnetic radiation emitted by certain stars that might fill large regions of space for long periods of time [11]. The systematic influence appears in the form of the rates of production or destruction of one enantiomer being larger than that of the other. The term $\sqrt{\varepsilon}f(t)$ represents random fluctuations with root-mean-square value $\sqrt{\varepsilon}$. Since the assumptions about rates of production for biomolecules, their catalytic activities and their concentrations determine the coefficients A , B and C , rather than the details of the chemical reaction scheme, the model is constrained by our general understanding of the prebiotic chemistry. Equation (19.3.17) provides a way to assess whether a given prebiotic model can produce and maintain the required asymmetry in a reasonable amount of time.

Furthermore, equation (19.3.17) can also give us a quantitative measure for the systematic chiral influence g . Detailed analysis, [16, 18] of this equation has

shown that the sensitivity of the bifurcation to the systematic influence depends on the rate at which the system moves through the critical point λ_c , i.e. we assume that $\lambda = \lambda_0 + \gamma t$, so that the initial value of λ_0 is less than λ_c , but that λ gradually increases to a value larger than λ_c at an average rate γ . This process may correspond, for example, to a slow increase in the concentrations of biomolecules in the oceans. It has been shown, [16, 18] that the probability P of the system making a symmetry-breaking transition to the asymmetric state favored by the systematic chiral influence g is given by

$$P = \frac{1}{2\pi} \int_{-\infty}^N e^{-x^2/2} dx \quad \text{where} \quad N = \frac{Cg}{\sqrt{\varepsilon/2}} \left(\frac{\pi}{B\gamma} \right)^{1/4} \quad (19.3.18)$$

Although derived in the context of biomolecular handedness, this formula is generally valid for any system that breaks a two-fold symmetry, such as mirror inversion. Using this formula, it is possible to understand the extraordinary sensitivity of bifurcation to small systematic biases that favor one enantiomer by increasing its production rate. For example, it can be estimated that the chiral asymmetry of the electroweak interaction can create differences of the order of one part in 10^{17} between the enantiomers. Application of the above theory shows that if the autocatalytic production rate of the chiral molecules is faster than the racemization rates, then for a period in the range 10^4 to 10^5 years, the enantiomer favored by the electroweak force will dominate [16]. For such a scenario, there is currently no experimental evidence to show us how chiral autocatalysis with the required properties can originate in prebiotic chiral molecules.

Many different scenarios have been suggested for the possible origins of biomolecular handedness. An extensive review can be found in the literature [19]. Note that, even if one is considering a process of chiral asymmetry generation after “life” arose, equations of the type (19.3.17) can still be used to describe the symmetry-breaking process, but this time the model will contain as “reactants” the self-replicating unit of life.

Appendix 19.1: Mathematica Codes

CODE A: MATHEMATICA CODE FOR SOLVING THE SET OF EQUATION (19.3.6) AND (19.3.7)

```
(* Chemical kinetics showing chiral symmetry breaking *)
k1f = 0.5; k1r = 0.1; k2f = 0.1; k2r = 0.2; k3f = 0.5; S = 0.5;
T = 0.5; Soln1 = NDSolve[{
  XL'[t] == k1f*S*T - k1r*XL[t] + k2f*S*T*XL[t]
  - k3f*XL[t]*XD[t],
  XD'[t] == k1f*S*T - k1r*XD[t] + k2f*S*T*XD[t]
  - k3*XL[t]*XD[t],
  XL[0] == 0.002, XD[0] == 0.0}, {XL, XD}, {t, 0, 100},
  MaxSteps -> 500]
```

The solution can be plotted using the following command:

```
Plot[Evaluate[{XL[t], XD[t]}/.Soln1], {t, 0, 100}]
```

